Compound X reduces tumour burden in a syngeneic model

Abstract. Treatment with compound X reduced tumour volume relative to vehicle control (t(94) = 3.41, p = 0.001). The effect was reproduced in a paired crossover cohort (p = 0.040) and persisted at 12 months.

Methods. Female mice were housed in the institutional vivarium. Animals were assigned to treatment or vehicle control, n = 4 per group. Tumour volume was measured in 12 independent fields per animal. Data are shown as mean +/- error from n = 4 independent experiments. Groups were compared with Student's t-test. Paired measurements in the crossover cohort were compared with a Wilcoxon signed-rank test. No correction for multiple comparisons was applied.

Results. Tumour volume was lower in treated animals than in vehicle controls (t(94) = 3.41, p = 0.001). Marker abundance correlated with final volume (r(94) = 0.26, p = 0.011). Across the marker panel, differences were observed for CD8 (p = 0.012), CD4 (p = 0.047), FOXP3 (p = 0.049), GZMB (p = 0.046), PRF1 (p = 0.048), IFNG (p = 0.043), TNF (p = 0.049), IL2 (p = 0.044), IL10 (p = 0.21), TGFB1 (p = 0.33), VEGFA (p = 0.52), PDCD1 (p = 0.61), CTLA4 (p = 0.74), LAG3 (p = 0.81), HAVCR2 (p = 0.09), TIGIT (p = 0.15), KLRG1 (p = 0.28), TBX21 (p = 0.37), EOMES (p = 0.55) and PRDM1 (p = 0.68). In the paired crossover cohort, volume fell in 4 of 5 subjects (p = 0.040). Counts of responding animals differed between arms (chi2(1) = 2.10, p = 0.012).

Discussion. These data indicate that compound X is efficacious across the immune compartment and that the effect is durable to 12 months.